Kapture Finance Collections Voicebot

High-Level Design (HLD) - AI Delivery Intern Take-Home

Prototype: Vapi + FastAPI + ngrok | Agent: Maya

1. Executive Summary

Maya is an outbound collections voice agent for Kapture Finance. The prototype verifies the intended customer before any debt disclosure, retrieves account details from a mock backend, captures a promise-to-pay, optionally sends a mock payment link, and records a final call disposition. Authentication is enforced by the backend using a generated auth token rather than by prompt discretion alone.

2. Architecture & Voice Pipeline

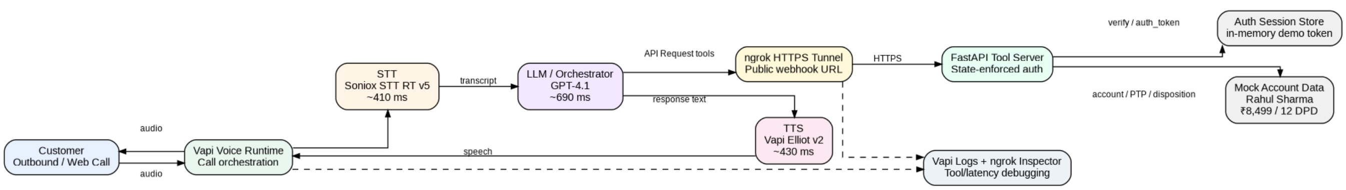

Figure 1. End-to-end voicebot architecture.

Hop	Component	Prototype Budget	Notes
1	STT	~410 ms	Soniox STT RT v5; streaming transcription
2	LLM / Orchestrator	~690 ms	GPT-4.1; intent, state, tool selection
3	Tool API	<500 ms target	FastAPI via ngrok; local mock endpoints
4	TTS	~430 ms	Vapi Elliot v2; professional voice
End-to-end	Turn response	~1.5-2.5 s target	Depends on tool calls/network

3. Conversation State Machine

State	Action	Lock / Transition Rule
START	Assistant introduces Maya/Kapture Finance and asks for Rahul Sharma.	No debt information.
UNVERIFIED	Collect full name, DOB, phone last four.	Debt/account tools are locked.
VERIFYING	Call verify_customer.	Only backend result can establish verification.
VERIFIED	Receive verified=true + auth_token.	Sensitive tools require exact auth_token.

ACCOUNT_DISCLOSURE	Call get_account_details and state tool-returned account facts.	No invented values.
INTENT	Classify will-pay / hardship / dispute / already-paid / DNC / wrong-person / callback / hostile.	Route to compliant branch.
PTP	Capture exact amount + date, call log_promise_to_pay.	Vague commitment is not a PTP.
PAYMENT_LINK	On consent, call send_payment_link.	Sending link is not payment confirmation.
DISPOSITION	Call mark_disposition exactly once for final outcome.	Avoid contradictory outcomes.
END	Short professional closing.	No new collection pressure.

4. Intents & Entities

Intent	Key Entities	Expected Action
will_pay	amount, ptp_date	Log PTP; optionally send payment link; mark promise_to_pay
cannot_pay / hardship	reason, possible date/amount	Empathize; do not invent plans; mark hardship
dispute	disputed item/reason	Do not argue; mark dispute; route/review
already_paid	payment date/reference if volunteered	Do not demand duplicate payment; mark already_paid
wrong_person	none	Reveal no debt; mark wrong_person; end
callback_requested	preferred date/time	Log callback_requested; do not overpromise
do_not_call	opt-out request	Stop collection discussion; mark do_not_call; end
hostile	none	Stay calm; mark hostile; end if needed

5. Tools / API Contracts

verify_customer - POST /verify-customer - validates full_name, date_of_birth, phone_last4. Returns verified, auth_token, message.

get_account_details - POST /account-details - requires auth_token. Returns customer_name, loan_type, overdue_emi, days_past_due.

log_promise_to_pay - POST /promise-to-pay - requires auth_token plus customer_name, amount, ptp_date.

send_payment_link - POST /send-payment-link - requires auth_token plus customer_name, phone_number, amount. Returns a mock payment link.

mark_disposition - POST /mark-disposition - logs final outcome using an enumerated disposition value.

Full request/response schemas are provided separately in tool_schemas.json.

6. Authentication & Data Safety

- Every call begins UNVERIFIED.
- Before verification, Maya must not reveal loan type, EMI amount, overdue status, days past due, or collection purpose to a third party.
- verify_customer validates three identity factors from the mock record: full name, DOB, and phone last four.
- Successful verification returns a generated auth_token.
- Sensitive backend endpoints require the token; therefore the LLM cannot bypass verification simply by changing conversational state.
- The prototype uses an in-memory session store. Production should bind sessions to Vapi call_id and use Redis or another durable store.
- No OTP, password, card number, CVV, or bank PIN is requested.

7. Guardrails & Compliance

- Identify the agent/company without revealing debt before verification.
- No threats, harassment, shaming, invented legal consequences, or fabricated account facts.
- Do-not-call requests stop the collection conversation immediately and are logged.
- Wrong-person/third-party answers receive no debt disclosure.
- Already-paid and dispute intents are acknowledged without demanding duplicate payment.
- Calling hours should be enforced by the outbound scheduler using the client's legal/compliance policy; exact jurisdictional hours were not specified in the assignment.
- If a tool fails, the assistant must not stay silently 'verifying' or claim success. It should state that the system could not complete the action and avoid protected disclosure.

8. Edge Cases

Case	Behavior
Already paid	Acknowledge, do not argue, mark already_paid, close.
Dispute	Do not pressure; mark dispute; route for review.
Do-not-call	Acknowledge immediately, mark do_not_call, end.
Wrong person	No debt disclosure, mark wrong_person, end.
Voicemail / no response	No sensitive disclosure; after timeout mark no_response.
Abusive / hostile	Stay calm; no retaliation; mark hostile; end if necessary.
Language switch EN/HI	Respond in the customer's language if understood; all privacy rules remain unchanged.

9. Escalation & Disposition

The prototype records outcomes through mark_disposition. A production version should add a human handoff tool for disputes requiring account investigation, vulnerable-customer/hardship cases, repeated tool failures, or customer requests for an agent.

Allowed dispositions: promise_to_pay, already_paid, do_not_call, wrong_person, dispute, hardship, callback_requested, hostile, no_response.

10. Observability & Metrics

Area	Track
Logs	call_id, tool name, request/response status, auth result, disposition, errors, silence/timeouts
Latency	STT, LLM, tool round-trip, TTS, total turn latency
Business	containment rate, PTP rate, payment-link acceptance, already-paid rate
Reliability	tool success rate, 4xx/5xx rate, drop rate, silence timeout rate
Compliance	pre-auth disclosure violations, DNC handling success, third-party disclosure violations

11. Model / Voice / Transcriber Rationale

Soniox STT RT v5 was used for responsive English streaming transcription. GPT-4.1 was selected for reliable instruction following and structured tool use during the prototype. Vapi Elliot v2 was selected for a natural, professional tone appropriate for a sensitive collections conversation. These choices favor reliability for the take-home demo; a production cost/latency study could evaluate cheaper alternatives.

12. Tested Demo Paths

- Path A - Successful Promise-to-Pay
    - Customer verifies successfully.
    - Account details retrieved only after auth.
    - Customer commits to ₹8,499 on a specific date.
    - PTP tool called.
    - Payment link offered/sent when accepted.
    - Final disposition logged as promise_to_pay.
- Path B - Already Paid
    - Customer verifies successfully.
    - Customer states the EMI was already paid.
    - Maya does not argue or request duplicate payment.
    - mark_disposition is called with already_paid.
    - Call closes politely.

13. Known Limitations & Next Steps

- In-memory auth sessions are demo-only and reset on server restart.
- ngrok free URL may change after restart.
- Payment link is mocked; no real payment transaction occurs.

- No production CRM, dialer policy engine, signed webhook verification, Redis, or human transfer is implemented.
- A production rollout should add automated Vapi simulations/evals, secrets management, call-hour enforcement, and bilingual regression tests.

14. Build/Debug Notes

The main failure observed during testing was a long silence after Maya said she would verify the customer. Vapi logs showed model activity but no account tool event. The fix was to strengthen mandatory tool-call rules, prevent filler claims such as 'I'll verify', add explicit tool-failure behavior, and validate actual API calls through the ngrok inspector. The ngrok agent was also upgraded from an unsupported v3.3.1 to v3.39.10.